

VIDIT RAJESH PRABHU

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EDUCATION

Master of Science in Computer Science and Engineering <i>University at Buffalo</i>	2024 – 2026 GPA: 3.6 / 4.0
Bachelor of Engineering in Electronics and Telecommunication Engineering <i>Don Bosco Institute of Technology, Mumbai</i>	2018 – 2022 GPA: 3.6 / 4.0

EXPERIENCE

FourSteps Solutions <i>ML/AI Engineering Intern</i>	Fremont, California <i>April 2026 – Present</i>
- Identified and resolved a production RBAC vulnerability in a FastAPI backend, redesigning the permission architecture to enforce least-privilege access across roles and closing a billing data exposure risk - Proposed and shipped a permission layer during code review that separated internal analytics from customer-facing access, preventing a potential data leak in a live PostgreSQL backed system - Built a pytest suite verifying the full permission matrix against live SQLAlchemy models, traced request flow across multiple Python services to identify authorization gaps, and documented secure access patterns for future backend changes	
Bluebird AI <i>Software Development Intern</i>	San Ramon, California <i>June 2025 – August 2025</i>
- Delivered a production voice assistant from scratch as sole backend engineer, integrating Google Speech-to-Text and Gemini LLM across voice and text input, LLM routing, database retrieval, and plain language response generation - Designed a hybrid LLM routing layer handling common queries through predefined templates and open ended queries through controlled fallback logic, driving out of context response rate from 1 in 6 queries to near zero - Cut average response latency by approximately 30% measured via log timing by rearchitecting API concurrency, restructuring inference execution, and deploying behind Nginx to sustain throughput under real load	
Tata Consultancy Services Ltd. <i>Assistant System Engineer</i>	Mumbai, India <i>June 2022 – May 2024</i>
- Contributed to a 15% team-wide gain in order processing efficiency by analyzing SQL data patterns and workflow bottlenecks across a Siebel CRM serving 3,000+ users - Built and configured CRM workflows, applets, and field level validations for order processing and billing flows, translating business requirements into backend CRM logic and UI updates - Maintained high system availability by diagnosing failed workflows, recovering stuck transactions, and resolving backend issues across order and billing systems	

PROJECTS

AlphaSignal – Financial News Sentiment & Market Intelligence Agent
Fine-tuned FinBERT on 11,931 samples achieving 88.3% accuracy, built a LangGraph agent with RAG retrieval, Redis caching for sub-100ms queries, and GitHub Actions CI/CD
Clairvoyix – Private GPU Powered AI Compute Platform
Built a GPU-powered ML platform for the AEC industry running BART, SAM2, and T5 RAG across Docker microservices, cutting manual effort by 40% with cost estimates within a 10% margin
Distributed In-Memory Cache System
Built a Redis-like distributed cache from scratch in Python with consistent hashing across 3 nodes, write replication, and heartbeat failover sustaining 4137 req/s at 100% hit rate and 10.9ms median GET latency across 10000 requests
Log Aggregation Service
Built a real-time log aggregation pipeline in Go with gRPC ingestion, Kafka message queue, REST query API, and Prometheus observability deployed on Kubernetes, sustaining 3178 req/s with 14.2ms median latency across 10000 requests

SKILLS

Languages: Python, C, C++, Java, SQL, Go, JavaScript
Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Keras, OpenCV, FastAPI, Flask, LangGraph, LlamaIndex, React, Node.js
Tools & Infrastructure: AWS (EC2, RDS), Docker, Kubernetes, Redis, PostgreSQL, Apache Kafka, Git, CI/CD, Prometheus, REST APIs, gRPC, LLMs, MLflow, Vector Databases